MIL-I-27552B(USAF)
AMENDMENT 2
25 September 1968
Superseding
AMENDMENT 1
10 August 1966

MILITARY SPECIFICATION

INDICATOR, TEMPERATURE, THERMOCOUPLE EHU-15A/A, POTENTIOMETER TYPE, WHITE LIGHTED

This amendment forms a part of Military Specification MIL-I-27552B(USAF) dated 17 November 1965.

Page 1, title: Delete and substitute: "INDICATOR, TEMPERATURE, THERMOCOUPLE EHU-15A/A, POTENTIOMETER TYPE, WHITE LIGHTED."

Page 1, paragraph 1.1: Delete and substitute:

"1.1 Scope. This specification covers the requirements for one type of white lighted, potentiometer-type, thermocouple temperature indicator, designated EHU-15A/A."

Page 1, paragraph 1.2: Delete.

Page 1, paragraph 2.1: Delete and substitute:

"2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-E-5400	Electronic Equipment Aircraft, General Specification for
MIL-W-5846	Wire, Electrical, Chromel and/or Alumel, Thermocouple
MIL-C-14806	Coating, Reflection Reducing, for Instrument Cover Glasses and Lighting Wedges
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification for

FSC 6685

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STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Specifications and Standards Order of Precedence for the Selection of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-785	Requirements for Reliability Program (for Systems and Equipments)
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MS3106	Connector, Plug, Electric, Straight
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33585	Pointers, Dial, Standard Design of Aircraft Instrument
MS33586	Metals, Definition of Dissimilar
MS33639	Cases, Instrument, Clamp-Mounted, Aircraft
MS33678	Connector, Receptacle, Electric, Integral Mounting

(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)"

Page 3, paragraph 3.5.4.1: Delete and substitute:

"3.5.4.1 Mean-time-between-failure (MTBF). The indicator shall have a minimum acceptable MTBF of 500 hours (contract specified MTBF of 750 hours) at a 90 percent confidence level."

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Page 3, paragraph 3.5.5: Delete and substitute:

"3.5.5 Maintainability. The following shall be provided for in the design of the indicator:

a. Optimum accessibility of components requiring maintenance

b. Rapid and positive identification of replaceable defective parts, assemblies, and components

c. Minimum number and types of tools and test equipment (special and standard) required to perform maintenance."

Page 5, paragraph 3.5.7: Delete and substitute:

"3.5.7 Cover glass coating. All reflecting surfaces shall be coated with a reflection-reducing coating that meets the requirements of MIL-C-14806 and that will withstand the environmental conditions specified herein."

Page 5, paragraph 3.6, items d and i: Delete and substitute:

"d. Case leakage not exceeding 0.1 micron cubic foot per hour

i. Withstand acceleration of 2 g's when vibrated at 5 to 500 cps".

Page 6, paragraph 3.6, item t, line 2: Delete "50" and substitute "48".

Page 6, paragraph 3.6.1: Delete and substitute:

"3.6.1 Electromagnetic interference. The indicator shall meet the electromagnetic interference requirements of MIL-STD-461 for class ID equipment."

Page 7, paragraph 3.7.2, second, third, and fourth sentences: Delete and substitute: "The filling medium shall be a mixture of 88 to 92 percent nitrogen and the remainder helium. The absolute pressure of the filling medium shall be 12 to 14.7 psi."

Page 8, paragraph 3.12, second sentence: Delete "MIL-D-70327" and substitute "MIL-STD-100".

Page 11, paragraph 4.5.2.2: Delete items n and q. Delete items j and m and substitute:

"j. Electromagnetic interference

m. Thermal shock and fogging".

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Page 11, add:

"4.5.2.4 Sampling plan C. Five indicators shall be selected at random from the first 15 on the contract or order and subjected to the reliability test specified in 4.6.23.2."

Page 12, paragraph 4.6.4: Delete and substitute:

"4.6.4 Sealing. The indicator shall be tested for leaks by means of a mass-spectrometer-type helium leak detector. The detected leak rate shall not exceed 0.1 micron cubic foot per hour at the specified filling pressure."

Page 13, paragraph 4.6.10, lines 3 and 4: Delete "MIL-E-5272, procedure XII, curve B" and substitute "MIL-STD-810, method 514, equipment category (b), curve A".

Page 16, paragraph 4.6.19: Delete and substitute:

"4.6.19 Electromagnetic interference. The indicator shall be subjected to electromagnetic interference tests in accordance with MIL-STD-462. The test methods shall be as specified in MIL-STD-461 for class ID equipment."

Page 16, paragraph 4.6.22: Delete the title and substitute: "Thermal shock and fogging". Between the second and third sentences, insert: "During the fourth and eighth cycles, at the end of the 30-minute soak in tap water at 85°C, the indicator shall be removed from the hot bath and within 4 seconds the face of the indicator shall be immersed in tape water at 5° ±2°C to a depth of 1/8 ±1/16 inch for a period of 1 minute. The indicator shall then be removed from the 5°C water and the cover glass immediately inspected for internal fogging or deposits."

Page 17, paragraph 4.6.23.1: Delete the fourth sentence and substitute: "Acceptance permitting production and the total test hours shall be determined in accordance with MIL-STD-781, test plan I." In the last line, delete "MIL-R-26667" and substitute "MIL-STD-781".

Page 18, paragraph 4.6.25, lines 3 and 4: Delete "MIL-E-5272, procedure I, for a period of 50 hours" and substitute "MIL-STD-810, method 509, for a period of 48 hours.

Page 18, paragraph 4.6.26: Delete.

Page 18, paragraph 5.1.1 through 5.2.3: Delete "MIL-P-7936" wherever referenced and substitute "MIL-STD-794."

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Page 18, paragraph 5.3: Delete and substitute:

"5.3 Marking for shipment. Interior packages and exterior shipping containers shall be marked for shipment in accordance with MIL-STD-129. The nomenclature shall be as follows:

Indicator, Temperature, Thermocouple EHU-15A/A Potentiometer Type, White Lighted."

Page 18, paragraph 6.1, line 1: Delete "The indicators" and substitute "The EHU-15A/A indicator".

Page 18, paragraph 6.2, item b: Delete.

Page 19, paragraph 6.3, last sentence: Delete "Systems Engineering Group, Attn: SEFIP" and substitute "Aeronautical Systems Division, Attn: ASNFI-20".

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